

Qualification Pack



Junior Instrumentation Technician (Process Control)

QP Code: IAS/Q3003

Version: 5.0

NSQF Level: 3

Qualification Pack

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IAS/Q3003: Junior Instrumentation Technician (Process Control)

Brief Job Description

Junior Instrumentation Technician (Process Control) carries out duties related to general site/plant readiness and usability and assists in preventive maintenance under supervision and guidance.

Personal Attributes

This job requires the individual to be disciplined, assertive, team player, possess analytical skills and problem-solving ability, effective communicator and have the ability to work under pressure.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
2. [IAS/N0103: Task Reporting - Process Control](#)
3. [IAS/N0105: Safety, Health and Environment Process Control](#)
4. [IAS/N0300: Site Readiness and Instrument Usability - Process Control](#)
5. [IAS/N0301: Preventive Maintenance Process Control](#)
6. [IAS/N9001: Work effectively with teams](#)

Qualification Pack (QP) Parameters

Sector	Instrumentation
Sub-Sector	Instrumentation & Automation
Occupation	Maintenance
Country	India
NSQF Level	3
Credits	11
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3111.9900

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Minimum Educational Qualification & Experience	8th grade pass with 1 Year of experience relevant field OR 8th grade pass with 1 year of (NTC/ NAC) after 8th
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	26/05/2025
NSQC Approval Date	26/05/2022
Version	5.0
Reference code on NQR	2022/ETR/IASC/06498
NQR Version	5

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

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IAS/N0103: Task Reporting - Process Control

Description

The OS unit is about reporting and record keeping as per company processes and job description for Instrumentation Technicians

Scope

The scope covers the following :

- This Unit Task covers the following:
- Reporting- faults
- Reporting PM
- Reporting Corrective Maintenance
- Reporting unusual occurrence
- Reporting Theft
- Reporting security breach

Elements and Performance Criteria

Report faults

To be competent, the user/individual on the job must be able to:

PC1. Report and Escalate faults/issues to immediate supervisor

Prepare PM Report

To be competent, the user/individual on the job must be able to:

PC2. Complete entry of preventive maintenance check lists/reports

Prepare Corrective Maintenance report

To be competent, the user/individual on the job must be able to:

PC3. Complete entry of Corrective Maintenance Check lists /reports

Report unusual occurrence

To be competent, the user/individual on the job must be able to:

PC4. Complete report on noticing any visible changes in control valve installation or its accessories.
Report for immediate attention of supervisor

Report theft

To be competent, the user/individual on the job must be able to:

PC5. Report any theft in control valve assembly/spares to supervisor

Report security breach

To be competent, the user/individual on the job must be able to:

PC6. Report suspicious movement of new persons near control valve installation to security and supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Understand basic company policy and maintenance strategy as applied to Control Valve
- KU2.** Understand importance of reporting potential failures during Control Valve Health visits / Preventive Maintenance checks/Observation to supervisor
- KU3.** Provide accurate time for jobs undertaken especially during opportunistic/annual shut down to supervisor
- KU4.** Provide status on critical spares consumption so that supervisor can take proactive steps
- KU5.** Prepare daily log and failure reports
- KU6.** Furnish basic data to supervisor related to specification of control valve
- KU7.** Send internal mails related to PM or Corrective maintenance to supervisor
- KU8.** Record Maintenance history

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write mails about work
- GS2.** Write reports and logs in company prescribed formats
- GS3.** Make notes about the observations in the plant and share with the supervisor and co-workers as appropriate
- GS4.** Write to management about feedback and unresolved issues
- GS5.** Read and comprehend formats and check lists for preventive and corrective maintenance.
- GS6.** Read and understand company policies
- GS7.** Communicate the issues / faults / corrective actions / warnings / suggestions with complete details to the supervisor and co-workers as appropriate
- GS8.** Make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS9.** How to prioritize jobs during multiple breakdown situation
- GS10.** Plan and organize work to achieve targets and deadlines
- GS11.** Discuss customer needs with co-workers and identify most appropriate solution
- GS12.** Can diagnose control valve faults and communicate effectively to Process and Instrumentation supervisor.
- GS13.** Use the existing information to arrive at actionable decision points
- GS14.** Use the existing information for improving the customer satisfaction
- GS15.** Use the existing information to optimize solution and company business
- GS16.** Analyze problems and identify causes and possible solutions
- GS17.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS18.** Anticipate problems, risks and opportunities and utilize these for mitigation and business optimization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Report faults</i>	5	5	-	-
PC1. Report and Escalate faults/issues to immediate supervisor	5	5	-	-
<i>Prepare PM Report</i>	10	15	-	-
PC2. Complete entry of preventive maintenance check lists/reports	10	15	-	-
<i>Prepare Corrective Maintenance report</i>	10	15	-	-
PC3. Complete entry of Corrective Maintenance Check lists /reports	10	15	-	-
<i>Report unusual occurrence</i>	10	10	-	-
PC4. Complete report on noticing any visible changes in control valve installation or its accessories. Report for immediate attention of supervisor	10	10	-	-
<i>Report theft</i>	5	5	-	-
PC5. Report any theft in control valve assembly/spares to supervisor	5	5	-	-
<i>Report security breach</i>	5	5	-	-
PC6. Report suspicious movement of new persons near control valve installation to security and supervisor	5	5	-	-
NOS Total	45	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	IAS/N0103
NOS Name	Task Reporting - Process Control
Sector	Instrumentation
Sub-Sector	Instrumentation & Automation
Occupation	Maintenance
NSQF Level	4
Credits	1
Version	3.0
Last Reviewed Date	20/12/2022
Next Review Date	20/12/2025
NSQC Clearance Date	20/12/2022

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IAS/N0105: Safety, Health and Environment Process Control

Description

The OS unit is about adhering to Safety, Health and Environment (SHE) norms in Process Plants.

Scope

The scope covers the following :

- The scope of this task includes following / performing: SHE Instructions SHE Personal Protection Directives Support SHE Audit

Elements and Performance Criteria

Follow SHE-Instructions

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret and follow formal Instructions from SHE Dept.
- PC2.** Participate in the prescribed drills including familiarization of personal protective equipment, fire extinguisher and first aid.
- PC3.** Follow Instructions on Work permit, Fire permit and Hazardous Area Classification, Fire and explosion hazards

Follow SHE for Personal Protection

To be competent, the user/individual on the job must be able to:

- PC4.** Use right personal protective equipment, such as: Use safety shoes and helmet at all times in plant Use ear muffs in prescribed high decibel areas Adapt right posture during lifting of heavy objects. Use correct gas mask/breathing air apparatus as per area and directives Use safety goggles and gloves as per directives

Support SHE - Audit

To be competent, the user/individual on the job must be able to:

- PC5.** Support supervisor during SHE Audit Check for leaks Check for unsafe scaffolding and temporary installations Check for correct storage of cleaning solvents and consumable such as waste cloth Ensure safe working in crane movement area. Check housekeeping including oil and water spillages on floor Check for improper bolting or open enclosures / junction boxes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understands SHE Instruction Understands Safety, Health and Environment directives Understands emergency procedures during site emergencies Understands Hazards in process as well as process media associated with these hazards
- KU2.** Demonstrates familiarity and understanding of documents related to SHE Understands SHE emergency procedures for the site. Understands Hazardous Area Classifications and Zones 0,1,2 Understands procedures of permits to work issued by Process personnel for maintenance work.

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Use Formats and check list for SHE
- GS2.** Write emails and messages about SHE related issues
- GS3.** Company's SHE policy.
- GS4.** Use of Formats and check list for SHE audits
- GS5.** Understands SHE Directives and SOPs related to maintenance work
- GS6.** Product literature and manuals relevant for the job
- GS7.** Company information about working practices at the site
- GS8.** Information displayed at the workplace
- GS9.** Describe site conditions and issues to co-workers and supervisor
- GS10.** Communicate to the management in meetings about Control Valve Health issues to get their support
- GS11.** Interact with coworkers and gather information related to process and site conditions
- GS12.** Make decisions pertaining to SHE in the concerned area of work
- GS13.** Ensures his personal protective equipment and arrangement of tools conform to safe working norms before he commences his /her daily tasks
- GS14.** Follows SOP/Vendor manual safety instructions with respect to dismantling, assembly and gland repair work on Control Valve
- GS15.** Discuss customer needs with co-workers and identify most appropriate solution
- GS16.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS17.** Identify immediate or temporary solutions to resolve delays
- GS18.** Discuss use the available information with co-workers to arrive at actionable decision points
- GS19.** Analyze problems in team and identify causes and possible solutions
- GS20.** Collaborate with co-workers to analyze, and evaluate the information gathered from collective observation, experience, reasoning, or communication, as a guide to teamwork

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow SHE-Instructions</i>	15	15	-	-
PC1. Interpret and follow formal Instructions from SHE Dept.	5	5	-	-
PC2. Participate in the prescribed drills including familiarization of personal protective equipment, fire extinguisher and first aid.	5	5	-	-
PC3. Follow Instructions on Work permit, Fire permit and Hazardous Area Classification, Fire and explosion hazards	5	5	-	-
<i>Follow SHE for Personal Protection</i>	5	5	-	-
PC4. Use right personal protective equipment, such as: Use safety shoes and helmet at all times in plant Use ear muffs in prescribed high decibel areas Adapt right posture during lifting of heavy objects. Use correct gas mask/breathing air apparatus as per area and directives Use safety goggles and gloves as per directives	5	5	-	-
<i>Support SHE - Audit</i>	5	5	-	-
PC5. Support supervisor during SHE Audit Check for leaks Check for unsafe scaffolding and temporary installations Check for correct storage of cleaning solvents and consumable such as waste cloth Ensure safe working in crane movement area. Check housekeeping including oil and water spillages on floor Check for improper bolting or open enclosures / junction boxes	5	5	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	IAS/N0105
NOS Name	Safety, Health and Environment Process Control
Sector	Instrumentation
Sub-Sector	Instrumentation & Automation
Occupation	Maintenance
NSQF Level	3
Credits	2
Version	3.0
Last Reviewed Date	20/12/2022
Next Review Date	20/12/2025
NSQC Clearance Date	20/12/2022

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IAS/N0300: Site Readiness and Instrument Usability - Process Control

Description

The OS unit is about maintaining general Site Readiness and Instrument Usability as applicable for process control industry

Scope

The scope covers the following :

- The scope of this task includes: Instructions relating to Process Plant Permits to work, Check Sheets & Formats Plant working and Instrumentation documents Field Instrumentation, control loops & Control valves On line Analyzers and Analyzer House DCS & PLC Package Units -Boilers, Heaters, Compressors, Chillers etc. Fire and Gas Detection system Site Guidelines of Field Installations - Visual, Integrity Checks & Electrical Safety norms Site Guidelines - House Keeping and Lighting Site Guidelines -Consumables, Execution and Follow up action

Elements and Performance Criteria

Observe Instructions relating to Process Plant

To be competent, the user/individual on the job must be able to:

- PC1.** Follow Process Plant related instructions and directives, including equipment, location, lay out, procedures, forms etc. covering: Process package units such as Boilers, compressors, heaters, chillers including equipment location /lay out etc. Work Permits Various formats and Check Sheets Field Instrumentation and Control loops Recognize and locate Control Valves Recognize and locate On line Analyzers and Analyzer House Recognize and locate DCS and PLC Recognize and locate Fire and Gas systems Safety, Health, Environment and emergency procedures

Follow Processes regarding Work Permits, Check Sheets & Formats

To be competent, the user/individual on the job must be able to:

- PC2.** Follow rules regarding various permits to work and use the right one depending on the job he/she undertakes. This includes: Normal working in plant WHITE permit Drilling, Grinding, Welding, Gas cutting in plant - RED Permit or Fire Permit. Excavation work for underground cables and pipe work etc.- EXCAVATION Permit
- PC3.** Understand the purpose of Check sheets used in Instrumentation Maintenance work and make entries in it: Site Hygiene Check Sheets Calibration Check Sheets Preventive Maintenance check sheets

Follow plant working and instrumentation documents

To be competent, the user/individual on the job must be able to:

- PC4.** Read and Use the Plant working document relevant to a particular Instrument tag/control loop he/she is working on, as required, relating to: Main Process Plants Package units such as Boilers, Heaters, Compressors Chillers etc.
- PC5.** Read and use the Instrumentation document relevant to a particular Instrument tag he/she is working with, as required: P and I diagram Instrument Index Data Sheets Hook Up diagram Loop diagram Instrument lay out diagram. Wiring Diagrams Cause and Effect diagram

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Support Servicing of Field Instrumentation and Control valves under guidance

To be competent, the user/individual on the job must be able to:

- PC6.** Attend to maintenance requests and resolve the problems with guidance. Request for supervisors advice and assistance during difficult problem resolution. This includes: Measurement sensors and associated impulse lines/sensor cables Pneumatic/electronic transmitters Controllers and control room receivers Control valves and other final control elements.
- PC7.** Perform basic overhaul and testing under supervision. Globe, Cage type, Butterfly, Rotary control valves Control Valve accessories, Solenoid Valves, Fail safe shut down devices Pneumatic and Digital Valve positioners Control Valve Data Sheet, control Valve characteristics i.e. quick opening, equal percentage, Linear Overhaul, testing and calibration of Control Valves

Perform routine checks for On line Analyzers and Analyzer House

To be competent, the user/individual on the job must be able to:

- PC8.** Carry out routine checks of: Analyzer sampling system Sensor and electronics section (liquid analyzers, Gas analyzers, Gas Chromatograph etc.) Receivers and accessories in Analyzer house. Calibration Gas cylinders storage and procedures.

Identify and locate DCS& PLC devices

To be competent, the user/individual on the job must be able to:

- PC9.** Locate the field devices and identifies the interface units able to work on the system with guidance.

Locate and identify trouble in PackageUnits Boilers, Heaters, Compressors, Chillers etc.

To be competent, the user/individual on the job must be able to:

- PC10.** Locate faults relating to operation of package units, location, relates instrument tags to respective Process package unit and undertakes routines and perform basic trouble shooting in these units.

Locate Fire and Gas Detection system and Maintain under guidance

To be competent, the user/individual on the job must be able to:

- PC11.** Refer to overview of fire and gas detection, Locate the main field devices, identify interface units and perform maintenance work on the system with guidance.

Follow Site Guidelines of Field Installations Visual , Integrity Checks and Electrical Safety norms

To be competent, the user/individual on the job must be able to:

- PC12.** .Check for visual damage, tampering: Check for visible damage to specified Instruments (in an inventory list or otherwise provided by the Supervisor) caused by impact of an external body.a. Check for prominent damage to Accessories of the specifiedInstruments caused by impact of an external body.b. Check for water ingress in indoor working area due to seepage/roof leaks /damaged windows.c. Check for misuse of installation by other agencies for example, using instrument installation as support or for scaffolding build up. d. Check for prominent deterioration due to environment for example, corrosion / proximity to hot surfaces / process leaks etc.

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- PC13.** Check for any visible loss of installation integrity. Integrity here means that the instrument and the installation is in a healthy working condition without any unusual appearance, breakage or foreign objects fallen in/on/around the installation which is likely to cause or lead to instrument malfunction.a. Check for improper closure of junction boxes, panels, cable termination.b. Check for improper tubing /impulse piping connectionc. Check for unauthorized /unexplained cable connection disconnection d. Check for unauthorized /unexplained process impulse lines /instrument air connection /disconnection.
- PC14.** Check for potential electrical problems due to deviation from standard electrical practicesa. Check for unused flood light /field plug connections with trailing cable. b.Check for bare, untagged or un-insulated wiresc. Check for any modifications or deviation in explosion proof enclosures and intrinsic safety installation with special focus on bolting integrityd. Check to ensure no deviation in ex proof enclosures and intrinsic safety installation with special focus on bolting integrity

Follow guidelines regarding House Keeping and Lighting

To be competent, the user/individual on the job must be able to:

- PC15.** Check for floor condition and cluttering of items Check for wet/slippery work areas, improper storage of items and cluttering of items on work areas. These are potential safety hazards
- PC16.** Check for lighting and their operation. Check for unsafe temporary wiring of lighting. Check in plant and indoor areas

Monitor and maintain Consumables and perform follow-up

To be competent, the user/individual on the job must be able to:

- PC17.** Check consumption and storage of consumables. Check for excessive consumption /visible wastage of cleaning solvents, lubrication oil and grease Check for excessive consumption of other related consumables such as waste cloth and gloves Check for proper storage of solvent and waste cloth (potential fire hazard).
- PC18.** Complete follow up action as per assigned areas of responsibility and stipulated instructions. Items found unusual outside this boundary to be reported to supervisor. Complete entry of relevant site Readiness check sheet and obtain required approval /endorsement. Replace protective covers which have been removed

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understands the purpose of the process plant and its operations.
- KU2.** Understands the layout of the plant and the location of various facilities.
- KU3.** Understands site conditions and how these impact the operation of instrumentation and control equipment.
- KU4.** Understands the organization and reporting structure
- KU5.** Understands rules and regulations to be followed under normal and emergency conditions
- KU6.** Understands working hours, shifts, off days and leave entitlements
- KU7.** Understands the job description and responsibility, if any.
- KU8.** Type of chemicals and other process material used in the organization and how these impact the site conditions
- KU9.** Hazardous chemicals and their handling procedures.

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- KU10.** Type of instruments and controllers used in the plant.
- KU11.** Type of instruments, accessories and their locations that falls under the individuals domain of work.
- KU12.** How instrumentation is located using plant layout drawings.
- KU13.** How instrumentation maintenance is performed in the organization.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Use Formats and check list for Site Readiness planning and reports
- GS2.** Write emails and messages about site related issues
- GS3.** Company policy related to site Readiness
- GS4.** Formats for site Readiness check sheets, list
- GS5.** Describe condition of instruments and accessories and issues to co- workers and supervisor
- GS6.** Communicate to the management in meetings about site issues which need management attention
- GS7.** Interact with coworkers and gather information related to process and instruments conditions
- GS8.** Make decisions about the site, in consultation with the Supervisor
- GS9.** Prioritize daily tasks to conduct site survey effectively
- GS10.** Understand real needs of the customer and suggest most appropriate solution
- GS11.** Support customer when they need help
- GS12.** Diagnoses reasons for down time due to instrument failure
- GS13.** Track recurring failures in instruments for reasons and concludes
- GS14.** Identify immediate or temporary solutions to resolve delays and discuss with the Supervisor
- GS15.** Use the existing information to arrive at actionable decision points
- GS16.** Use the existing information for improving the customer satisfaction
- GS17.** Use the existing information to optimize solution and company business
- GS18.** Analyze problems and identify causes and possible solutions
- GS19.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS20.** Anticipate problems, risks and opportunities and utilize these for mitigation and business optimization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Observe Instructions relating to Process Plant</i>	10	10	-	-
PC1. Follow Process Plant related instructions and directives, including equipment, location, lay out, procedures, forms etc. covering: Process package units such as Boilers, compressors, heaters, chillers including equipment location /lay out etc. Work Permits Various formats and Check Sheets Field Instrumentation and Control loops Recognize and locate Control Valves Recognize and locate On line Analyzers and Analyzer House Recognize and locate DCS and PLC Recognize and locate Fire and Gas systems Safety, Health, Environment and emergency procedures	10	10	-	-
<i>Follow Processes regarding Work Permits, Check Sheets & Formats</i>	10	10	-	-
PC2. Follow rules regarding various permits to work and use the right one depending on the job he/she undertakes. This includes: Normal working in plant WHITE permit Drilling, Grinding, Welding, Gas cutting in plant - RED Permit or Fire Permit. Excavation work for underground cables and pipe work etc.- EXCAVATION Permit	5	5	-	-
PC3. Understand the purpose of Check sheets used in Instrumentation Maintenance work and make entries in it: Site Hygiene Check Sheets Calibration Check Sheets Preventive Maintenance check sheets	5	5	-	-
<i>Follow plant working and instrumentation documents</i>	10	10	-	-
PC4. Read and Use the Plant working document relevant to a particular Instrument tag/control loop he/she is working on, as required, relating to: Main Process Plants Package units such as Boilers, Heaters, Compressors Chillers etc.	5	5	-	-
PC5. Read and use the Instrumentation document relevant to a particular Instrument tag he/she is working with, as required: P and I diagram Instrument Index Data Sheets Hook Up diagram Loop diagram Instrument lay out diagram. Wiring Diagrams Cause and Effect diagram	5	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Support Servicing of Field Instrumentation and Control valves under guidance</i>	10	20	-	-
PC6. Attend to maintenance requests and resolve the problems with guidance. Request for supervisors advice and assistance during difficult problem resolution. This includes: Measurement sensors and associated impulse lines/sensor cables Pneumatic/electronic transmitters Controllers and control room receivers Control valves and other final control elements.	5	15	-	-
PC7. Perform basic overhaul and testing under supervision. Globe, Cage type, Butterfly, Rotary control valves Control Valve accessories, Solenoid Valves, Fail safe shut down devices Pneumatic and Digital Valve positioners Control Valve Data Sheet, control Valve characteristics i.e. quick opening, equal percentage, Linear Overhaul, testing and calibration of Control Valves	5	5	-	-
<i>Perform routine checks for On line Analyzers and Analyzer House</i>	5	5	-	-
PC8. Carry out routine checks of: Analyzer sampling system Sensor and electronics section (liquid analyzers, Gas analyzers, Gas Chromatograph etc.) Receivers and accessories in Analyzer house. Calibration Gas cylinders storage and procedures.	5	5	-	-
<i>Identify and locate DCS& PLC devices</i>	5	5	-	-
PC9. Locate the field devices and identifies the interface units able to work on the system with guidance.	5	5	-	-
<i>Locate and identify trouble in PackageUnits Boilers, Heaters, Compressors, Chillers etc.</i>	5	5	-	-
PC10. Locate faults relating to operation of package units, location, relates instrument tags to respective Process package unit and undertakes routines and perform basic trouble shooting in these units.	5	5	-	-
<i>Locate Fire and Gas Detection system and Maintain under guidance</i>	5	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Refer to overview of fire and gas detection, Locate the main field devices, identify interface units and perform maintenance work on the system with guidance.	5	5	-	-
<i>Follow Site Guidelines of Field Installations Visual , Integrity Checks and Electrical Safety norms</i>	15	25	-	-
PC12. .Check for visual damage, tampering: Check for visible damage to specified Instruments (in an inventory list or otherwise provided by the Supervisor) caused by impact of an external body.a. Check for prominent damage to Accessories of the specifiedInstruments caused by impact of an external body.b. Check for water ingress in indoor working area due to seepage/roof leaks /damaged windows.c. Check for misuse of installation by other agencies for example, using instrument installation as support or for scaffolding build up. d. Check for prominent deterioration due to environment for example, corrosion / proximity to hot surfaces / process leaks etc.	5	10	-	-
PC13. Check for any visible loss of installation integrity. Integrity here means that the instrument and the installation is in a healthy working condition without any unusual appearance, breakage or foreign objects fallen in/on/around the installation which is likely to cause or lead to instrument malfunction.a. Check for improper closure of junction boxes, panels, cable termination.b. Check for improper tubing /impulse piping connectionc. Check for unauthorized /unexplained cable connection disconnection d. Check for unauthorized /unexplained process impulse lines /instrument air connection /disconnection.	5	10	-	-
PC14. Check for potential electrical problems due to deviation from standard electrical practicesa. Check for unused flood light /field plug connections with trailing cable. b.Check for bare, untagged or un-insulated wiresc. Check for any modifications or deviation in explosion proof enclosures and intrinsic safety installation with special focus on bolting integrityd. Check to ensure no deviation in ex proof enclosures and intrinsic safety installation with special focus on bolting integrity	5	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow guidelines regarding House Keeping and Lighting</i>	7	6	-	-
PC15. Check for floor condition and cluttering of items Check for wet/slippery work areas, improper storage of items and cluttering of items on work areas. These are potential safety hazards	5	3	-	-
PC16. Check for lighting and their operation. Check for unsafe temporary wiring of lighting. Check in plant and indoor areas	2	3	-	-
<i>Monitor and maintain Consumables and perform follow-up</i>	7	10	-	-
PC17. Check consumption and storage of consumables. Check for excessive consumption /visible wastage of cleaning solvents, lubrication oil and grease Check for excessive consumption of other related consumables such as waste cloth and gloves Check for proper storage of solvent and waste cloth (potential fire hazard).	2	5	-	-
PC18. Complete follow up action as per assigned areas of responsibility and stipulated instructions. Items found unusual outside this boundary to be reported to supervisor. Complete entry of relevant site Readiness check sheet and obtain required approval /endorsement. Replace protective covers which have been removed	5	5	-	-
NOS Total	89	111	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IAS/N0300
NOS Name	Site Readiness and Instrument Usability - Process Control
Sector	Instrumentation
Sub-Sector	Instrumentation & Automation
Occupation	Maintenance
NSQF Level	3
Credits	2
Version	3.0
Last Reviewed Date	NA
Next Review Date	26/05/2025
NSQC Clearance Date	26/05/2022

Qualification Pack

IAS/N0301: Preventive Maintenance Process Control

Description

The OS unit is about performing Preventive Maintenance for general instrumentation in Process Controls under guidance.

Scope

The scope covers the following :

- This Unit Task covers the following : Obtaining PM-work permit. Organizing PM-Preparatory tasks Executing PM-Schedule Preparing PM-Process List PM-Planning Performing PM-Visual Checks & action Completion of PM schedule. Preparing PRM-List Listing PM to be taken during Shut down Following up consolidated PM list

Elements and Performance Criteria

Obtain PM-work permit

To be competent, the user/individual on the job must be able to:

- PC1.** Obtain work permit (mandatory) from the Process supervisor before commencing maintenance work.

Organize PM- Preparatory tasks

To be competent, the user/individual on the job must be able to:

- PC2.** Organize for the days Preventive Maintenance task which include: Check sheets and related documents Tools and tackles and calibrating equipment Personal protective safety equipment

Execute PM Schedule

To be competent, the user/individual on the job must be able to:

- PC3.** Execute Preventive maintenance jobs as per available Preventive Maintenance Schedule.

Prepare PM-Process List

To be competent, the user/individual on the job must be able to:

- PC4.** Prepare Process list from process supervisor.

Plan PM Schedule

To be competent, the user/individual on the job must be able to:

- PC5.** Plan for next days preventive maintenance schedule

Perform PM Visual Checks & Actions

To be competent, the user/individual on the job must be able to:

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- PC6.** Carry out Visual Checks & takes corrective actions wherever possible or else transfer job to shut down list. Check abnormal noise & locates source. Check for abnormal vibration locates source- corrects /informs supervisor. Check for gland leak and arrests .If not possible informs supervisor and transfers to shut down list Check for Instrument air leaks arrests leak-if not possible informs supervisor and transfers to shut down list Check for control valve bonnet and body flange leaks-informs Process supervisor& Instrument supervisor Check for poor electric integrity like exposed terminal and opens enclosure- rectifies. If not permitted informs supervisor& transfer to shut down list Check pneumatic and electric connections to solenoid valve Check for pneumatic connections in Pneumatic transmitters and or I/Pconverters Check for integrity of valve positioners feedback link/connection and rectifies fault after informing Process supervisor Check for environment impact, water ingress/corrosion rectifies fault.Or else .transfers to shut down list. Check for loose bolting and rattling in control valve and accessories including manual operator- rectifies fault

Complete PM schedule

To be competent, the user/individual on the job must be able to:

- PC7.** Complete preventive maintenance schedule list of Field Instrumentation and controls. Close the list.

Prepare PRM List

To be competent, the user/individual on the job must be able to:

- PC8.** Collect and consolidate daily-diagnostic messages from control valves which have a digital valve controller and Smart Transmitters and record the same in either Preventive Maintenance list or Opportunistic shut down list for execution.

List PMs to be performed during Shut down

To be competent, the user/individual on the job must be able to:

- PC9.** Include preventive maintenance jobs during annual shut down or opportunistic shut down list.

Follow up PM consolidated list

To be competent, the user/individual on the job must be able to:

- PC10.** Follow up on consolidated preventive maintenance list and closes list.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understands PM (Preventive Maintenance) norms as defined by the company
- KU2.** Understands Production targets and production loss figures for the month and contribution by control valves
- KU3.** Aware of Maintenance Policy of the company with respect to control valve maintenance strategy
- KU4.** Trouble Shooting of: Control valve body including valve movement and glands. Actuator section and valve positioners and diagnostic messages from Digital Valve controllerA. Control valve accessoriesB. Solenoid valve and Limit switches
- KU5.** Hazardous area zone classification and process media hazards
- KU6.** Use of Control Valve Manual when required
- KU7.** PST, a feature of DCS

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Use Formats and check list for Site Readiness planning and reports
- GS2.** Write emails and messages about site related issues
- GS3.** Company policy related to Preventive Maintenance
- GS4.** Monthly down time and resulting production loss
- GS5.** Use of Work Permit system
- GS6.** Formats for Preventive Maintenance check sheets, predictive maintenance list
- GS7.** Describe condition of control valves and accessories and issues to co- workers and supervisor
- GS8.** Communicate to the management in meetings about maintenance issues which need management attention
- GS9.** Interact with coworkers and gather information related to process and control valve conditions
- GS10.** Make decisions about timing and extent of preventive maintenance, in consultation with the Supervisor
- GS11.** Prioritize daily tasks to conduct Preventive Maintenance effectively
- GS12.** Understand real needs of the customer and suggest most appropriate solution
- GS13.** support customer when they need help
- GS14.** Diagnoses reasons for down time due to control valve failure
- GS15.** Track recurring failures in control valves and analyses reasons and concludes
- GS16.** Identify immediate or temporary solutions to resolve delays and discuss with the Supervisor
- GS17.** Use the existing information to arrive at actionable decision points
- GS18.** Use the existing information for improving the customer satisfaction
- GS19.** Use the existing information to optimize solution and company business
- GS20.** Analyze problems and identify causes and possible solutions
- GS21.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS22.** Anticipate problems, risks and opportunities and utilize these for mitigation and business optimization

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Obtain PM-work permit</i>	5	5	-	-
PC1. Obtain work permit (mandatory) from the Process supervisor before commencing maintenance work.	5	5	-	-
<i>Organize PM- Preparatory tasks</i>	10	30	-	-
PC2. Organize for the days Preventive Maintenance task which include: Check sheets and related documents Tools and tackles and calibrating equipment Personal protective safety equipment	10	30	-	-
<i>Execute PM Schedule</i>	10	10	-	-
PC3. Execute Preventive maintenance jobs as per available Preventive Maintenance Schedule.	10	10	-	-
<i>Prepare PM-Process List</i>	10	10	-	-
PC4. Prepare Process list from process supervisor.	10	10	-	-
<i>Plan PM Schedule</i>	-	-	-	-
PC5. Plan for next days preventive maintenance schedule	-	-	-	-
<i>Perform PM Visual Checks & Actions</i>	10	20	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC6. Carry out Visual Checks & takes corrective actions wherever possible or else transfer job to shut down list. Check abnormal noise & locates source. Check for abnormal vibration locates source- corrects /informs supervisor. Check for gland leak and arrests .If not possible informs supervisor and transfers to shut down list Check for Instrument air leaks arrests leak-if not possible informs supervisor and transfers to shut down list Check for control valve bonnet and body flange leaks-informs Process supervisor& Instrument supervisor Check for poor electric integrity like exposed terminal and opens enclosure- rectifies. If not permitted informs supervisor& transfer to shut down list Check pneumatic and electric connections to solenoid valve Check for pneumatic connections in Pneumatic transmitters and or I/Pconverters Check for integrity of valve positioners feedback link/connection and rectifies fault after informing Process supervisor Check for environment impact, water ingress/corrosion rectifies fault.Or else .transfers to shut down list. Check for loose bolting and rattling in control valve and accessories including manual operator- rectifies fault	10	20	-	-
<i>Complete PM schedule</i>	5	15	-	-
PC7. Complete preventive maintenance schedule list of Field Instrumentation and controls. Close the list.	5	15	-	-
<i>Prepare PRM List</i>	7	8	-	-
PC8. Collect and consolidate daily-diagnostic messages from control valves which have a digital valve controller and Smart Transmitters and record the same in either Preventive Maintenance list or Opportunistic shut down list for execution.	7	8	-	-
<i>List PMs to be performed during Shut down</i>	10	15	-	-
PC9. Include preventive maintenance jobs during annual shut down or opportunistic shut down list.	10	15	-	-
<i>Follow up PM consolidated list</i>	5	15	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Follow up on consolidated preventive maintenance list and closes list.	5	15	-	-
NOS Total	72	128	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IAS/N0301
NOS Name	Preventive Maintenance Process Control
Sector	Instrumentation
Sub-Sector	Instrumentation & Automation
Occupation	Maintenance
NSQF Level	3
Credits	2
Version	3.0
Last Reviewed Date	NA
Next Review Date	26/05/2025
NSQC Clearance Date	26/05/2022

Qualification Pack

IAS/N9001: Work effectively with teams

Description

This NOS unit is about building relationships and working with people and groups inside and outside the organization, using skills and habits, to achieve the team goals and objectives.

Scope

The scope covers the following :

- Work as per organisational team environment
- Communicate effectively
- Co-operate with team members and superiors
- Respect customs / preferences and gender / ability differences "

Elements and Performance Criteria

Work as per the organisational team environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify team objectives and goals, team members by name, their role and responsibilities, greet them appropriately and respond to their greetings
- PC2.** comply with organisation's policies and procedures for working with team members within and outside the organisation—especially related to privacy, confidentiality and security
- PC3.** work as per the environment to build trust and mutual respect
- PC4.** participate in decision making by providing facts and figures, give / accept constructive suggestions, take initiatives to help team members arrive at workable decisions and meet the goals
- PC5.** accept decisions professionally and support even if they do not match suggestions and personal views

Communicate effectively

To be competent, the user/individual on the job must be able to:

- PC6.** communicate professionally as per organisation's protocols, using appropriate mode of communication—verbal, written, mail, phone or text—and clearly articulate the message to ensure that the receiver understands the message
- PC7.** listen to team members attentively, respond promptly, seek / provide clarifications if required
- PC8.** share important information with the team timely and refrain from overloading them with unnecessary and unsolicited information

Co-operate with team members and superiors

To be competent, the user/individual on the job must be able to:

- PC9.** perform own role, receive inputs from others and make adjustments within permissible rules as per requirement, to produce output in time for other team members to follow

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- PC10.** help team members to perform their role effectively and provide any clarifications/support they need, including tools /equipment / common resources as well as resolve any contentious issues amicably, involving the team lead or the supervisor if needed
- PC11.** let team members know in good time if commitments cannot be carried out, explaining the reasons, and provide alternate solutions, if any; let the team lead know about this
- PC12.** act in the interest of the team and the organisation, take initiative to correct the wrong, seek help or escalate if needed to ensure that things do not 'fall through the gap' and team goals are achieved

Respect customs / preferences and gender / ability differences

To be competent, the user/individual on the job must be able to:

- PC13.** follow organisation's policies and statutory guidelines w.r.t seeking information about others' customs / preferences, making references or comments on social customs / preferences, and refrain from hurting sentiments
- PC14.** accommodate team members' preferences to the extent feasible, and in case they come in the way of fulfilling team goals, discuss with the supervisor/ team leader
- PC15.** ensure personal behaviour, conduct and communication styles, taking gender and disability of the person into consideration
- PC16.** list the different types of disabilities with their respective issues and ways to help them overcome challenges
- PC17.** use inclusive language, verbal as well as non-verbal, irrespective of the disability and the gender of the person
- PC18.** ensure equal treatment for all clients, colleagues and co-workers while respecting their personal space

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organisation's policies on dress code , workplace behaviour , performance management,incentives,delivery standards,information security,etc.
- KU2.** Organisation's hierarchy and escalation matrix
- KU3.** importance of the individual's role in the workflow
- KU4.** work area inspection procedures and practices
- KU5.** different types of information that colleagues might need and the importance of providing this information when it is required
- KU6.** deeper understanding of actions and consequences of gender based behaviour
- KU7.** knowledge of gender based concepts, issues and legislation
- KU8.** organisation standards and guidelines to be followed for PwD and knowledge about laws, acts and provisions defined for PwD by the statutory bodies and the right way to use them including various medical conditions associated with PwD
- KU9.** health and safety requirements at workplace for PwD and information about various institutes working for PwD to enable in providing livelihood opportunities for PwD
- KU10.** rights and duties at workplace with respect to PwD and various government / private schemes and benefits available for PwD

Qualification Pack

KU11. process of recruiting people for a particular job profile w.r.t PwD and gender including rights and duties at workplace with respect to gender sensitivity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete forms such as work orders, invoices and maintenance records
- GS2.** fill up appropriate forms, activity logs and attendance sheets as per the organisation's format in English and/or local language
- GS3.** write basic accident or incident report as witnessed in an appropriate format to the relevant authority
- GS4.** read warnings, instructions and other text material on product labels, components, etc.
- GS5.** read relevant signage, warnings, labels or descriptions on equipment, etc. while carrying out work activities
- GS6.** listen effectively and orally communicate information
- GS7.** ask for clarification and advice from the concerned person
- GS8.** make decisions on a suitable course of action or response keeping in view resource utilisation while meeting
- GS9.** plan and organise work to achieve targets and deadlines
- GS10.** understand needs of the customer, suggest most appropriate solution and support them whenever needed
- GS11.** match symptoms of the fault noticed to the cause of the problem
- GS12.** anticipate and avoid hazards that may occur during repairs because of tools, materials used or repair processes
- GS13.** spot process disruptions and delays
- GS14.** practice and acceptance of gender and its concepts
- GS15.** develop empathy across genders and towards PwD
- GS16.** reflect on own gender identity, gender roles and PwD issues
- GS17.** engage and participate in discussions to end gender and disability discrimination
- GS18.** improve and modify work practices
- GS19.** maintain positive and effective relationships with colleagues and customers

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work as per the organisational team environment</i>	15	8	-	5
PC1. identify team objectives and goals, team members by name, their role and responsibilities, greet them appropriately and respond to their greetings	4	4	-	-
PC2. comply with organisation's policies and procedures for working with team members within and outside the organisation—especially related to privacy, confidentiality and security	4	-	-	2
PC3. work as per the environment to build trust and mutual respect	2	-	-	1
PC4. participate in decision making by providing facts and figures, give / accept constructive suggestions, take initiatives to help team members arrive at workable decisions and meet the goals	4	4	-	1
PC5. accept decisions professionally and support even if they do not match suggestions and personal views	1	-	-	1
<i>Communicate effectively</i>	6	10	-	1
PC6. communicate professionally as per organisation's protocols, using appropriate mode of communication—verbal, written, mail, phone or text—and clearly articulate the message to ensure that the receiver understands the message	2	6	-	1
PC7. listen to team members attentively, respond promptly, seek / provide clarifications if required	2	-	-	-
PC8. share important information with the team timely and refrain from overloading them with unnecessary and unsolicited information	2	4	-	-
<i>Co-operate with team members and superiors</i>	8	18	-	1
PC9. perform own role, receive inputs from others and make adjustments within permissible rules as per requirement, to produce output in time for other team members to follow	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. help team members to perform their role effectively and provide any clarifications/support they need, including tools /equipment / common resources as well as resolve any contentious issues amicably, involving the team lead or the supervisor if needed	-	6	-	1
PC11. let team members know in good time if commitments cannot be carried out, explaining the reasons, and provide alternate solutions, if any; let the team lead know about this	2	-	-	-
PC12. act in the interest of the team and the organisation, take initiative to correct the wrong, seek help or escalate if needed to ensure that things do not 'fall through the gap' and team goals are achieved	4	6	-	-
<i>Respect customs / preferences and gender / ability differences</i>	11	14	-	3
PC13. follow organisation's policies and statutory guidelines w.r.t seeking information about others' customs / preferences, making references or comments on social customs / preferences, and refrain from hurting sentiments	2	4	-	-
PC14. accommodate team members' preferences to the extent feasible, and in case they come in the way of fulfilling team goals, discuss with the supervisor/ team leader	2	-	-	1
PC15. ensure personal behaviour, conduct and communication styles, taking gender and disability of the person into consideration	2	6	-	1
PC16. list the different types of disabilities with their respective issues and ways to help them overcome challenges	1	-	-	1
PC17. use inclusive language, verbal as well as non-verbal, irrespective of the disability and the gender of the person	2	4	-	-
PC18. ensure equal treatment for all clients, colleagues and co-workers while respecting their personal space	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	50	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IAS/N9001
NOS Name	Work effectively with teams
Sector	Instrumentation
Sub-Sector	Instrumentation & Automation
Occupation	Generic
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2031
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Qualification Pack

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	0	0	50	10
IAS/N0103.Task Reporting - Process Control	45	55	-	-	100	20
IAS/N0105.Safety, Health and Environment Process Control	25	25	-	-	50	20
IAS/N0300.Site Readiness and Instrument Usability - Process Control	89	111	-	-	200	20
IAS/N0301.Preventive Maintenance Process Control	72	128	-	-	200	20
IAS/N9001.Work effectively with teams	40	50	-	10	100	10
Total	291	399	-	10	700	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.